The role of exercise in the treatment of obesity.

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The United States is in the midst of a significant public health problem that relates to obesity and inactivity. This epidemic has far-ranging consequences for our workforce and our children and shows no signs of slowing in the near future. Significant research has been performed on the effects of exercise for the reduction of body weight; results of most studies indicate that exercise alone has a small effect on body-weight reduction independent of caloric restriction. However, when combined with dietary restriction, exercise has a synergistic effect and enhances weight loss beyond the effect of diet alone. In addition, exercise has been shown to have significant beneficial effects on cardiovascular and metabolic risk factors independent of actual weight loss, and losing just a small amount of weight can have a significant beneficial effect on these parameters. Genetic factors related to obesity have been found to be positively modified when persons incorporate physical activity into their lifestyle. Sitting time appears to be an independent risk factor for the development of metabolic risk factors; persons who spend more time sitting and watching television have worse metabolic profiles, even if they achieve the recommended amount of physical activity per week, than do those who move about throughout the day. Exercise also is essential for the prevention of weight gain over a life span, although the amount required to prevent weight gain may be closer to twice the amount of exercise recommended by the current Physical Activity Guidelines for Americans (www.health.gov/paguidelines). In many ways, the physiatrist is the most well prepared of all the specialists to address the complex, multidimensional problems of obesity and inactivity.